Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Distributed Systems Architecture, Agentic AI Workflows, LLM Orchestration, Autonomous Anomaly Detection, Machine Learning & Forecasting (XGBoost, SARIMA), Data Lakehouse Design, Dimensional Modeling (Star/Snowflake, SCD I/II), Data Quality Governance & Lineage, Unit Economics & Predictive Analytics

Programming Languages & Tools: Python, PySpark, Apache Spark, AWS (Redshift, S3, Glue, Kinesis), Databricks, Snowflake, Azure (ADF, Synapse), KIRO IDE, Power BI (DAX), Tableau, Collibra, Git, Scikit-Learn, SQL

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Software Dev Engineer II - AI & Profit Intelligence Platforms — **May 2022 – Present**

- **Architected agentic AI workflows and prompt orchestration** utilizing KIRO IDE, GitHub Copilot, and custom scripts to automate schema discovery, regex classification, and code migration, accelerating delivery turnaround by 40-50%.
- **Engineered autonomous ML anomaly detection systems** operating over distributed Spark and Cloud data pipelines to identify schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream BI integrity.
- **Built high-throughput cloud lakehouse architectures** using AWS (S3, Redshift), PySpark, Databricks, and Snowflake, crunching 100s of TBs of transactional data with dimensional modeling (Star/Snowflake, SCD 1/2) under 99.9% availability SLAs.
- **Deployed scalable predictive and time-series models** including XGBoost, Random Forests, and SARIMA to forecast field support capacity and customer behavior, cutting operational response times by 20%.
- **Developed automated testing frameworks for Conversational AI** evaluating virtual assistant platforms (SpectrumIQ) through automated test pipelines, slashing manual QA turnaround time by 50%.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Designed and deployed machine learning pipelines** leveraging Scikit-Learn, PySpark, and XGBoost on multi-terabyte customer datasets to extract actionable operational signals and boost service reliability by 20%.
- **Architected time-series forecasting frameworks** using SARIMA algorithms to predict field support demand and truck rolls, enabling optimized workforce capacity allocation.
- **Implemented MLOps and automated data profiling** integrating automated feature extraction and data drift monitoring to ensure continuous model performance in production.

Business Intelligence & Data Governance Specialist — **May 2022 – Present**

- **Engineered automated data quality firewalls** and Collibra source-to-target data lineage, cutting operational governance and data reconciliation turnaround by over 80%.
- **Delivered executive decision intelligence platforms** authoring Key Operating Metric (KOM) semantic models and interactive dashboards in Power BI (DAX) and Tableau to drive corporate strategy.
- **Optimized dimensional schemas and data pipelines** across Azure Synapse, SQL, and Snowflake, enforcing strict data governance and high-availability standards.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.